Xiaoyi (Estelle) Wang

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Education

University of Toronto Toronto, ON
Honours Bachelor of Science, Computer Science Major | Neuroscience Major, CGPA 3.96 *Sep 2023 – Apr 2027*

- **Relevant Coursework:**
  - *ML & AI:* Machine Learning, Artificial Intelligence, Probabilistic Learning and Reasoning, Visual Computing, Natural Language Computing, Image Understanding
  - *Computer Science:* Computer Organization, Software Design, Theory of Computation, Data Structures and Algorithm Design, Databases, Operating Systems
  - *Math & Statistics:* Linear Algebra, Calculus, Probability Statistics and Data Analysis, Linear Programming
  - *Life Sciences:* Lab in Neuroscience, Neurogenomics, Molecular and Cell Biology, Biochemistry, Human Physiology
- **Awards and honors:**
  - Dean's List Scholar 2024 & 2025 & 2026
  - New College Council In-Course Scholarship 2023–2024 & 2024–2025

Research Interests

Machine learning (ML) for healthcare; ML for women's health; Time-series data modeling; Wearable device; Computational neuroscience; Safe AI.

Research Experience

Modular Representation Learning for Wearable Time Series (CSC495) Sep 2026 – Present
Goldenberg Lab, undergraduate student researcher

- Perform literature review on time-series foundational models and human activity recognition (HAR) classification models.
- Perform exploratory data analysis on HAR datasets PAMAP2, WISDM, and CAPTURE24.
- Propose and test non-linear GVARs to fit time-series signals and use Student-t distributions to fit non-Gaussian, tail-heavy residuals.
- Create a per-activity baseline for model evaluation.

ML: Menstrual Cycle-Informed Predictions from Wearable Data (CSC494) May 2026 – Aug 2026
Student-initiated project, primary student researcher

- Conducted a systematic literature review on factors that might be influenced by the menstrual cycle: cognition, emotion, motivation, workplace productivity, and sleep.
- Performed feature engineering and data quality assessment for data smoothing, missing data handling, duplicate detection, and temporal aggregation.
- Built a data preprocessing pipeline that integrated, analyzed, and imputed data from the mcPHASES dataset.
- Used supervised ML models (Random Forest, CatBoost, Linear Regression, etc.), cross-validation, and ablation studies to analyze the predictiveness of physiological and hormonal features for 3 distinct targets: fatigue, stress, and deep-sleep.

Presentations

ML: Menstrual Cycle-Informed Predictions from Wearable Data Aug 25, 2026
Poster presentation, Undergraduate Summer Research Poster Showcase

Projects

Painting Classification Model Jan 2026 – Apr 2026

- Implemented an ML pipeline for painting classification, achieved 89% accuracy.
- Performed data preprocessing on numerical, ordinal, and text (TF–IDF) features.
- Explored various model families: kNN, Logistic Regression, Linear SVM, Random Forest, Neural Networks.

Taxi Database Design Oct 2025 – Nov 2025

- Designed a relational database using SQL DDL from natural language specifications.
- Wrote & tested complex SQL (PostgreSQL) queries: joins, subqueries, aggregation.
- Implemented embedded SQL in Python (psycopg2) to automate querying, data extraction, and data analysis.

Software Design (Card Game Software) Oct 2024 – Dec 2024

- Collaborated in an Agile team to deliver a production-style application following Clean Architecture.
- Implemented Use Cases in Java, translating user and system requirements into technical design.
- Adopted GitHub, Jira-style workflows, and peer reviews to support collaboration and team delivery.

Work Experience

Heritage Lab (HMB491) Remote
AI Data Analyst & Research Intern *Sep 2026 – Present*

- Research and document 500+ patient-doctor conversations for translation model training.
- Create English-Inuktitut translation corpus for over 1450 sentence/word pairs, as of Oct 29, 2026.
- Train translation AI model for medical topics, promoting indigenous healthcare.

KITE Research Institute – University Health Network (UHN) Remote
Technical Support Student *Sep 2026 – Present*

- Annotate joints in video frames using DeepLabCut 3.0.2 for joint hypermobility detection.
- Augment training images with various lighting, background, and subject features such as tattoos to debias model.
- Upgrade modeling backbone for faster, lighter-weight model training and prediction.

University of Toronto (UofT) Remote
Biotechnology and Computer Science Content Editor *Jun 2025 – Apr 2026*

- Delivered accessible data science and machine learning materials to 200+ life science students per term.
- Troubleshot and standardized course content for 3 undergraduate courses at UofT (HMB201, HMB301, HMB491), improving clarity, technical accuracy, and consistency.
- Collaborated on Jupyter and GitHub for cross-platform teamwork.

Skills

- **Programming:** Python, Java, C, SQL, Assembly (proficient); JavaScript, C#, R (familiar).
- **Machine Learning:** Scikit-Learn, Numpy, Pandas, PyTorch, SciPy, Matplotlib.
- **Software Development:** Compilers, Linux/Unix OS, GIT, OOP, REST APIs.
- **Languages:** English (fluent); Mandarin (fluent); French (beginner).
- **Laboratory:** Western Blotting, EEG, cell culture, RT-qPCR, RNA-seq, DNA isolation.